Code No: 118BJ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, December - 2020
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. Explain about the top down and bottom up design approach of an embedded system design. [15]

2. With a neat block diagram, describe the architecture of 8051 micro controller. [15]

3.a) Explain the interrupt handling mechanism of 8051.
b) Explain about logical instructions with examples. [5+10]

4.a) Explain in detail about Assembly Language Programming Tools and Techniques.
b) With examples, explore use of JUMP and CALL instructions. [11+4]

5. What are the various modes of serial data communication? Interface UART with 8051μc. Also draw the block diagram of UART. [15]

6.a) Explain how intelligent LCD displays can be interfaced with 8051 Microcontroller?
b) With example, explain message queues and mail boxes. [7+8]

7.a) Discuss various methods adopted to reduce power consumption in RTOS.
b) Explain Real time scheduling consideration in RTOS. [7+8]

8.a) Draw ARM7 Architecture and explain its features in detail.
b) Give the basic functions of CAN bus. [10+5]

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